

ETHICAL HACKING TASK-1

PASSIVE AND ACTIVE RECONNAISSANCE REPORT

1. OSINT REPORT

Introduction:

Open Source Intelligence (OSINT) refers to the process of collecting information from publicly available sources without directly interacting with the target system. It is the first step in ethical hacking and helps in gathering useful information.

Objective:

The objective of this report is to gather publicly available information about the target domain using passive reconnaissance techniques.

Methodology:

Information was collected using publicly available tools and search engines without sending direct requests to the target system.

Tools Used:

WHOIS, DNS Lookup, Shodan, theHarvester, Google Dorking

Information Collected:

- Domain Name: internship-corp.local
- DNS Records: A Record, MX Record, NS Record
- Emails and subdomains using theHarvester
- Publicly available data using search engines and Shodan

Analysis:

The collected data provides an understanding of the target's structure and possible entry points such as exposed services and email patterns.

Conclusion: OSINT is an important phase in reconnaissance as it helps gather valuable information without interacting with the target system.

2. SCAN SUMMARY

Introduction:

Active scanning is used to identify live hosts, open ports, and running services in a network by directly interacting with the target system.

Objective:

The objective is to perform network scanning and identify active systems and services.

Tools Used:

Nmap, Netdiscover

Methodology:

Network scanning was performed using Nmap to detect open ports and identify services running on the target system.

Command Used:

```
nmap -p 21,22,80 scanme.nmap.org
```

Sample Output:

PORT	STATE	SERVICE
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21/tcp	closed	ftp
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22/tcp	open	ssh
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80/tcp	open	http
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Host Discovery:

Live hosts in the network were identified using scanning techniques.

Service Enumeration:

Services running on open ports were identified for further analysis.

Analysis:

The scan results show that multiple ports are open and services such as FTP, SSH, and HTTP are running on the system.

Commands used:

```
C:\Users\NAVEEN>nmap -p 21,22,80 scanme.nmap.org
```

Nmap Scan output:

```
C:\Users\NAVEEN>nmap -p 21,22,80 scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-05 12:27 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.11s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT      STATE SERVICE
21/tcp    closed ftp
22/tcp    open  ssh
80/tcp    open  http
```

Conclusion:

Open ports may lead to vulnerabilities if not properly secured. Proper configuration is required to protect the system.

3. FINAL REPORT

Summary:

This project involved both passive and active reconnaissance techniques to gather and analyze information about a target system.

Work Done:

- Collected information using OSINT techniques
- Performed network scanning using Nmap
- Identified open ports and running services

Findings:

The scan results revealed that several ports were open on the system. Common services such as FTP, SSH, and HTTP were identified, indicating active network services.

Risks:

Open ports and running services can be potential entry points for attackers if they are not properly secured. These vulnerabilities may lead to unauthorized access or data breaches.

Recommendations:

- Close unused ports
- Keep all services updated
- Use firewalls to restrict access
- Monitor network activity regularly

Conclusion:

The reconnaissance process helps in identifying potential vulnerabilities in a system. Implementing proper security measures can protect the network from possible cyber threats and attacks.